Grokking Dynamic Programming Patterns for Coding Interviews

Master Dynamic Programming by learning coding patterns. Act smartly, and follow the Dynamic Programming patterns.

Level: Advanced | Study Time: 71h | Lessons: 73 | Playgrounds : 142

39,106 (Students)

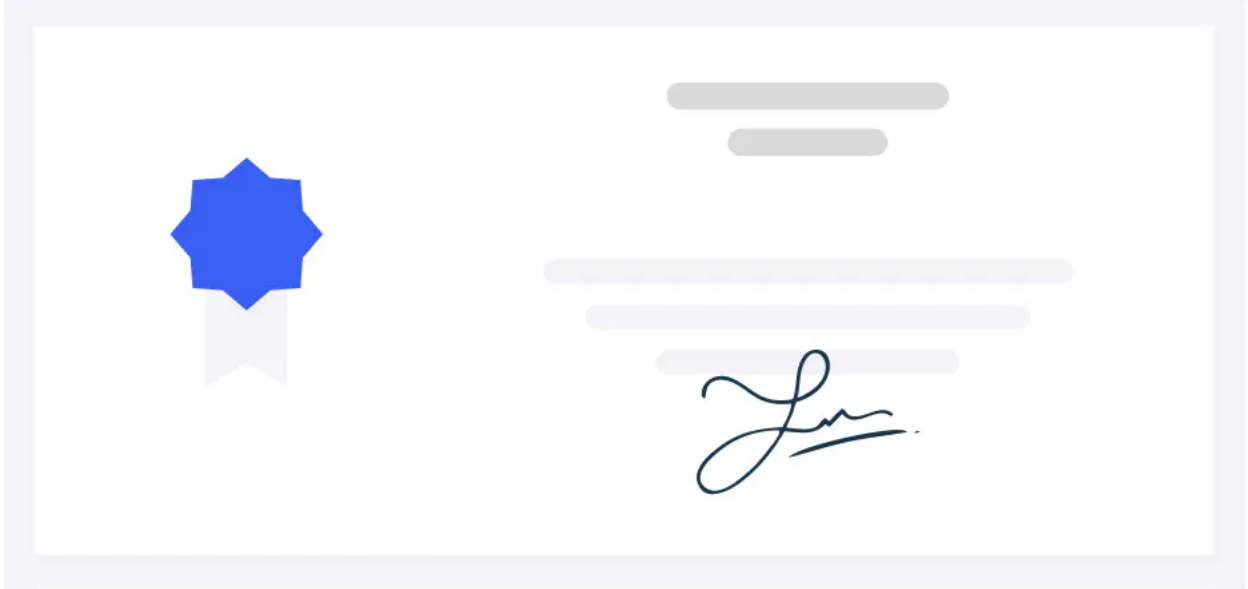
Course Overview

The fact is, Dynamic Programming (DP) problems can be some of the most intimidating on a coding interview. Even when it's actually clear if a problem can be solved using DP (which it rarely is), it can be pretty challenging to even know where to start on the solution. Unless, that is, you're trained on the approach to solving DP problems.

This course was made from scratch with just that goal in mind. It'll equip you with a set of easy-to-understand techniques to handle any DP problem. Rather than just having you try to memorize solutions, you'll be walked through five underlying DP patterns that can then be applied to solve 35+ DP problems. In each pattern, we'll start with a recursive brute-force solution -- this is actually the best way to start solving a DP problem! Once we have a recursive solution, we'll then apply the advanced DP methods of Memoization and Tabulation. The practice problems in this course were carefully chosen, covering the most frequently asked DP questions in coding interviews.



Start Reading



Certification on completing the course.

Download Certificate

Course Content

What is Dynamic Programming?

2. Pattern 1: 0/1 Knapsack — 13 chapters

Introduction
0/1 Knapsack
Solution: 0/1 Knapsack
Equal Subset Sum Partition
Solution: Equal Subset Sum Partition
Subset Sum
Solution: Subset Sum
Minimum Subset Sum Difference
Solution: Minimum Subset Sum Difference
Count of Subset Sum
Solution: Count of Subset Sum
Target Sum
Solution: Target Sum

3. Pattern 2: Unbounded Knapsack — 10 chapters

Unbounded Knapsack
Solution: Unbounded Knapsack
Rod Cutting
Solution: Rod Cutting
Coin Change
Solution: Coin Change
Minimum Coin Change
Solution: Minimum Coin Change
Maximum Ribbon Cut
Solution: Maximum Ribbon Cut

4. Pattern 3: Fibonacci Numbers — 12 chapters

Fibonacci numbers
Solution: Fibonacci numbers
Staircase
Solution: Staircase
Number factors
Solution: Number factors

Solution: Minimum jumps to reach the end

Minimum jumps with fee

Solution: Minimum jumps with fee

House thief

Solution: House thief

5. Pattern 4: Palindromic Subsequence

10 chapters

Longest Palindromic Subsequence

Solution: Longest Palindromic Subsequence

Longest Palindromic Substring

Solution: Longest Palindromic Substring

Count of Palindromic Substrings

Solution: Count of Palindromic Substrings

Minimum Deletions in a String to make it a Palindrome

Solution: Minimum Deletions in a String to make it a Palindrome

Palindromic Partitioning

Solution: Palindromic Partitioning

6. Pattern 5: Longest Common Substring

26 chapters

Longest Common Substring

Solution: Longest Common Substring

Longest Common Subsequence

Solution: Longest Common Subsequence

Minimum Deletions & Insertions to Transform a String into another

Solution: Minimum Deletions & Insertions to Transform a String into another

Longest Increasing Subsequence

Solution: Longest Increasing Subsequence

Maximum Sum Increasing Subsequence

Solution: Maximum Sum Increasing Subsequence

Shortest Common Super-sequence

Solution: Shortest Common Super-sequence

Minimum Deletions to Make a Sequence Sorted

Solution: Minimum Deletions to Make a Sequence Sorted

Longest Repeating Subsequence

Solution: Longest Repeating Subsequence

Subsequence Pattern Matching

Longest Bitonic Subsequence
Solution: Longest Bitonic Subsequence
Longest Alternating Subsequence
Solution: Longest Alternating Subsequence
Edit Distance
Solution: Edit Distance
Strings Interleaving
Solution: Strings Interleaving

7. Appendix 1 chapter

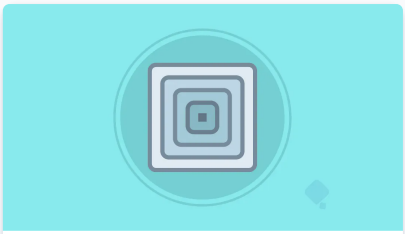
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Related Courses



Grokking the Coding Interview: Patterns for Coding Questions

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Grokking the Art of Recursion for Coding Interviews

Start Reading

What people say about our courses

MO JAFRI
Software Engineer

The courses which have "grokking" before them, are exceptionally well put together! These courses magically condense 3 years of CS in short bite-size courses and lectures (I have tried System Design, OODI, and Coding patterns). The Grokking courses are godsent, to be honest.

AHMET HANIF
Software Engineer

Whoever put this together, you folks are life savers. Thank you :)

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